

16.1) 식 (16.8)을 증명하라.

$$\sum_{i=1}^n h_{ij} = \sum_j h_{ij} = 1$$

$$h_{ij} = e_i^T H e_j = (e_i^T H) (H^T e_j)$$

$$h_{ij} = e_i^T H e_j = [e_i^T H] [H^T e_j] = [h_{i1} \dots h_{in}] [h_{1j} \dots h_{nj}]^T = \sum_{k=1}^n h_{ik} \cdot h_{kj}$$

$$\begin{aligned} \sum_{i=1}^n h_{ij} &= \mathbf{1}^T H \cdot e_j = \mathbf{1}^T (\Pi_{\mathbf{1}} + \Pi_{X_{\perp}}) e_j \\ &= \mathbf{1}^T \cdot \Pi_{\mathbf{1}} e_j \\ &= \mathbf{1}^T e_j \\ &= 1 \end{aligned}$$

식 (16.11)을 증명하라.

$$h_{ii} = \frac{1}{n} + (x_i - \bar{x})' (\mathbf{X}' \mathbf{X})^{-1} (x_i - \bar{x}) \quad (16.11)$$

으로 표현된다. 여기서 $x_i^T = (x_{i1}, \dots, x_{ik})$ (i^{th} column), $\bar{x} = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k)$, $\bar{x}_i = \sum_{j=1}^n x_{ij}/n$ 이고 $\mathbf{X}' \mathbf{X}$ 는

$$\mathbf{X}' \mathbf{X} = \begin{bmatrix} x_{11} - \bar{x}_1 & x_{21} - \bar{x}_1 & \dots & x_{k1} - \bar{x}_1 \\ x_{12} - \bar{x}_2 & x_{22} - \bar{x}_2 & \dots & x_{k2} - \bar{x}_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} - \bar{x}_1 & x_{2n} - \bar{x}_2 & \dots & x_{kn} - \bar{x}_k \end{bmatrix}$$

을 나타낸다.

$$\begin{aligned} h_{ii} &= e_i^T H e_i = \mathbf{1}^T X (X'X)^{-1} X' e_i \\ &= e_i^T (\Pi_{\mathbf{1}} + \Pi_{X_{\perp}}) e_i \\ &= e_i^T \Pi_{\mathbf{1}} e_i + e_i^T \Pi_{X_{\perp}} (X_{\perp}' X_{\perp})^{-1} X_{\perp}' e_i \\ &= \frac{1}{n} + (x_i - \mathbf{1} \bar{x})' (X_{\perp}' X_{\perp})^{-1} (x_i - \mathbf{1} \bar{x}) \\ &= \frac{1}{n} + (x_i - \mathbf{1} \bar{x})' (\mathbf{X}' \mathbf{X})^{-1} (x_i - \mathbf{1} \bar{x}) \end{aligned}$$

16.3) 식 (16.15)을 증명하라.

$$S^2(i) = \left[(n-k-1) S^2 - \frac{e_i^2}{1-h_{ii}} \right] / (n-k-2) \quad (16.15) \quad \text{공식 외우기}$$

$S(i)$: externally studentized residual

$$S_i^2 = \frac{1}{n-k-2} \sum_{j \neq i}^n (y_j - \hat{y}_{j(i)})^2, \quad y_{j(i)} = z_j^T \hat{\beta}_{(i)}, \quad \hat{\beta}_{(i)} = (X_{(i)}^T X_{(i)})^{-1} X_{(i)}^T y_{(i)}$$

$z_j^T = [z_{j1}, \dots, z_{jk}]$
↳ j^{th} row vector

$$\hat{\beta}_{(i)} = (X_{(i)}' X_{(i)})^{-1} X_{(i)}' y_{(i)} = \{X'X - z_i z_i^T\}^{-1} \{X'y - z_i y\}$$

$$X_{(i)}' X_{(i)} = X'X - z_i z_i^T, \quad X_{(i)}' y_{(i)} = X'y - z_i y$$

$$\begin{aligned} \{X'X - z_i z_i^T\}^{-1} &= (X'X)^{-1} + \frac{(X'X)^{-1} z_i z_i^T (X'X)^{-1}}{1 - z_i^T X'X z_i} \\ &= (X'X)^{-1} + \frac{(X'X)^{-1} z_i z_i^T (X'X)^{-1}}{1 - h_{ii}} \end{aligned}$$

$$\begin{aligned} (X'X - z_i z_i^T)^{-1} (X'y - z_i y) &= \left\{ (X'X)^{-1} + \frac{(X'X)^{-1} z_i z_i^T (X'X)^{-1}}{1 - h_{ii}} \right\} (X'y - z_i y) \\ &= (X'X)^{-1} X'y + \frac{(X'X)^{-1} z_i z_i^T (X'X)^{-1} X'y}{1 - h_{ii}} - (X'X)^{-1} z_i y - \frac{(X'X)^{-1} z_i z_i^T (X'X)^{-1} z_i y}{1 - h_{ii}} \\ &= \hat{\beta} + \frac{(X'X)^{-1} z_i z_i^T \hat{\beta}}{1 - h_{ii}} - (X'X)^{-1} z_i y - \frac{(X'X)^{-1} z_i h_{ii} y}{1 - h_{ii}} \\ &= \hat{\beta} + \frac{1}{1 - h_{ii}} \left\{ (X'X)^{-1} z_i z_i^T \hat{\beta} - (1 - h_{ii})(X'X)^{-1} z_i y - (X'X)^{-1} z_i h_{ii} y \right\} \\ &= \hat{\beta} + \frac{1}{1 - h_{ii}} \left\{ (X'X)^{-1} z_i (y_i - y_i) \right\} \\ &= \hat{\beta} - \frac{(X'X)^{-1} z_i}{1 - h_{ii}} \hat{e}_i \rightarrow \text{외우기} \end{aligned}$$

$$S_i^2 = \left[(n-k-1) S^2 - \frac{e_i^2}{1-h_{ii}} \right] / (n-k-2)$$

$$\begin{aligned} (n-k-2) S_i^2 &= \sum_{j \neq i}^n (y_j - \hat{y}_{j(i)})^2 = \sum_{j=1}^n (y_j - \hat{y}_{j(i)})^2 - \{y_i - \hat{y}_{i(i)}\}^2 \\ &= \sum_{j=1}^n \left(\hat{e}_j - \frac{\hat{e}_j h_{ji}}{1 - h_{ii}} \right)^2 - \left(\hat{e}_i + \frac{\hat{e}_i h_{ii}}{1 - h_{ii}} \right)^2 \\ &= \sum_{j=1}^n \left\{ \hat{e}_j^2 - 2 \cdot \frac{\hat{e}_j \hat{e}_i h_{ji}}{1 - h_{ii}} + \frac{(\hat{e}_i h_{ji})^2}{(1 - h_{ii})^2} \right\} - \frac{\hat{e}_i^2}{(1 - h_{ii})^2} \\ &= \sum_{j=1}^n \hat{e}_j^2 + \frac{\hat{e}_i^2 h_{ii}}{(1 - h_{ii})^2} - \frac{\hat{e}_i^2}{(1 - h_{ii})^2} \quad \text{goes to zero} \\ &= (n-k-1) \hat{\sigma}^2 - \frac{\hat{e}_i^2}{1 - h_{ii}} \end{aligned}$$



16.4) 식 (16.17) 을 증명하라.

$$t_i = r_i \left(\frac{n-k-2}{n-k-1-r_i^2} \right)^{\frac{1}{2}} \quad \dots (16.17)$$

$$t_i = \frac{y_i - \tilde{y}_i}{S(i) \sqrt{1 + x_i' [X(i)' X(i)]^{-1} x_i}} \quad \dots (16.16)$$

* 만약 y_i 가 이상치라면 $E[y_i - \tilde{y}_i] \neq 0$ 이 될 것이고, 이상치가 아니라면 $E[y_i - \tilde{y}_i] = 0$ 이 될 것이다.

* y_i 가 이상치가 아니라면 $t_i = \frac{y_i - \tilde{y}_i}{S(i) \sqrt{1 + x_i' [X(i)' X(i)]^{-1} x_i}}$ 은 자유도 $n-k-2$ 인 t 분포이다.

$$r_i = \frac{\hat{\epsilon}_i}{S \sqrt{1 - h_{ii}}} \quad \dots (16.13) \text{ (internally) Studentized residual}$$

$$r_i^* = \frac{\hat{\epsilon}_i}{S(i) \sqrt{1 - h_{ii}}} \quad \dots (16.14) \text{ (externally) Studentized residual}$$

$$S^2(i) = \left[(n-k-1)S^2 - \frac{e_i^2}{1-h_{ii}} \right] / (n-k-2) \quad \dots (16.15), e_i = \hat{\epsilon}_i$$

$$t_i = \frac{y_i - \tilde{y}_i}{S(i) \sqrt{1 + x_i' [X(i)' X(i)]^{-1} x_i}} \quad \dots (16.16)$$

$$t_i^2 = \frac{y_i - \tilde{y}_i}{S(i)^2 (1 + x_i' [X(i)' X(i)]^{-1} x_i)}$$

$$r_i^2 = \frac{e_i^2}{S^2(1-h_{ii})} \Rightarrow S^2 r_i^2 = \frac{e_i^2}{1-h_{ii}} \quad \therefore S^2(i) = \left[(n-k-1)S^2 - S^2 r_i^2 \right] / (n-k-2) \\ = \left[(n-k-1-r_i^2)S^2 \right] / (n-k-2)$$

$$\begin{aligned} [X(i)' X(i)]^{-1} &= [X'X - x_i x_i']^{-1} \\ &= (X'X)^{-1} + \frac{[X'X]^{-1} x_i x_i' [X'X]^{-1}}{1 - x_i' [X'X]^{-1} x_i} = (X'X)^{-1} + \frac{[X'X]^{-1} x_i x_i' [X'X]^{-1}}{1 - h_{ii}} \\ x_i' [X(i)' X(i)]^{-1} x_i &= x_i' (X'X)^{-1} x_i + \frac{x_i' [X'X]^{-1} x_i x_i' [X'X]^{-1} x_i}{1 - h_{ii}} \\ &= h_{ii} + \frac{h_{ii}^2}{1 - h_{ii}} = \frac{h_{ii} - h_{ii}^2 + h_{ii}^2}{1 - h_{ii}} = \frac{h_{ii}}{1 - h_{ii}} \end{aligned}$$

$$1 + \frac{h_{ii}}{1 - h_{ii}} = \frac{1}{1 - h_{ii}}$$

$$\therefore 1 + x_i' [X(i)' X(i)]^{-1} x_i = (1 - h_{ii})^{-1}$$

$$t_i^2 = (y_i - \tilde{y}_i)^2 \cdot \frac{n-k-2}{(n-k-1-r_i^2)S^2} \cdot (1 - h_{ii})$$

$$\begin{aligned} y_i - \tilde{y}_i &= y_i - x_i' \hat{\beta}(i) = y_i - x_i' \left(\hat{\beta} - \frac{\hat{\epsilon}_i}{1 - h_{ii}} (X'X)^{-1} x_i \right) \quad (i=1, \dots, n) \\ &= y_i - \hat{y}_i + \frac{\hat{\epsilon}_i h_{ii}}{1 - h_{ii}} = e_i + \frac{e_i h_{ii}}{1 - h_{ii}} = \frac{e_i}{1 - h_{ii}} \end{aligned}$$

$$y_i - \tilde{y}_i = \frac{e_i}{1 - h_{ii}}$$

$$\begin{aligned} \therefore t_i^2 &= \left(\frac{e_i}{1 - h_{ii}} \right)^2 \cdot \frac{n-k-2}{(n-k-1-r_i^2)S^2} \cdot (1 - h_{ii}) \\ &= \frac{n-k-2}{(n-k-1-r_i^2)} \cdot \frac{e_i^2}{S^2(1-h_{ii})} = \frac{n-k-2}{(n-k-1-r_i^2)} r_i^2 \end{aligned}$$

$$\therefore t_i = \left\{ \frac{n-k-2}{n-k-1-r_i^2} \right\}^{\frac{1}{2}} r_i$$

$$16.5) Cook \text{의 통계량에서 } \frac{(b - b(i))' (X'X)^{-1} (b - b(i))}{(k+1)S^2} = \frac{1}{k+1} \cdot \frac{h_{ii} e_i^2}{S^2(1-h_{ii})^2}$$

$$b - b(i) = \hat{\beta} - \hat{\beta}(i) = \hat{\beta} - (X(i)' X(i))^{-1} X(i)' y(i)$$

$$\begin{aligned} [X(i)' X(i)]^{-1} &= (X'X - x_i x_i')^{-1} = (X'X)^{-1} + \frac{[X'X]^{-1} x_i x_i' [X'X]^{-1}}{1 - x_i' [X'X]^{-1} x_i} \\ &= (X'X)^{-1} + \frac{[X'X]^{-1} x_i x_i' [X'X]^{-1}}{1 - h_{ii}} \end{aligned}$$

$$X(i)' y(i) = X' y - x_i' y_i$$

$$\begin{aligned} [X(i)' X(i)]^{-1} X(i)' y(i) &= \left\{ [X'X]^{-1} + \frac{[X'X]^{-1} x_i x_i' [X'X]^{-1}}{1 - h_{ii}} \right\} (X' y - x_i y_i) \\ &= \hat{\beta} + \frac{[X'X]^{-1} x_i x_i' \hat{\beta}}{1 - h_{ii}} - (X'X)^{-1} x_i y_i - \frac{[X'X]^{-1} x_i x_i' h_{ii} y_i}{1 - h_{ii}} \\ &= \hat{\beta} + (X'X)^{-1} x_i \frac{1}{1 - h_{ii}} \left\{ \tilde{y}_i - (1 - h_{ii}) y_i - h_{ii} y_i \right\} \\ &= \hat{\beta} + (X'X)^{-1} x_i \frac{\hat{\epsilon}_i}{1 - h_{ii}} \end{aligned}$$

$$\hat{\beta} - \hat{\beta}(i) = (X'X)^{-1} x_i \frac{\hat{\epsilon}_i}{1 - h_{ii}}$$

$$X(\hat{\beta} - \hat{\beta}(i)) = X(X'X)^{-1} x_i \frac{\hat{\epsilon}_i}{1 - h_{ii}}$$

$$X(\hat{\beta} - \hat{\beta}(i)) = \left(\frac{\hat{\epsilon}_i}{1 - h_{ii}} \right)^2 x_i' (X'X)^{-1} x_i = \frac{e_i^2}{(1 - h_{ii})^2} h_{ii}$$

$$\therefore \text{Cook's Statistic} = \frac{e_i^2 h_{ii}}{(1 - h_{ii})^2} \cdot \frac{1}{S^2(k+1)}$$

16.4 좀 더 빠르게 증명하기

$$\text{WTS: } r_i^* = r_i \left(\frac{n-k-2}{n-k-1-r_i^2} \right)^{\frac{1}{2}}$$

$$r_i^*: \text{externally studentized residual} = \hat{\epsilon}_i / \left\{ S(i) (1 - h_{ii}) \right\}^{\frac{1}{2}}$$

$$\Leftrightarrow S(i)^2 = \frac{\hat{\epsilon}_i^2}{(r_i^*)^2 (1 - h_{ii})}$$

$$\text{we know that } S(i)^2 = \left[(n-k-1)S^2 - \frac{e_i^2}{1-h_{ii}} \right] / (n-k-2) \rightarrow \text{from Q16.3}$$

$$\therefore \left[(n-k-1)S^2 - \frac{\hat{\epsilon}_i^2}{1-h_{ii}} \right] / (n-k-2) = \frac{\hat{\epsilon}_i^2}{(r_i^*)^2 (1-h_{ii})}$$

$$\left[(n-k-1)S^2(1-h_{ii}) - \hat{\epsilon}_i^2 \right] / (n-k-2) = \frac{\hat{\epsilon}_i^2}{(r_i^*)^2}$$

$$\left[(n-k-1) \frac{S^2(1-h_{ii})}{\hat{\epsilon}_i^2} - 1 \right] / (n-k-2) = \frac{1}{(r_i^*)^2}$$

$$= \left[(n-k-1) r_i^2 - 1 \right] / (n-k-2) = (r_i^*)^{-2}$$

$$\therefore r_i^{*2} = \frac{n-k-2}{(n-k-1)r_i^2 - 1} = r_i^2 \cdot \frac{n-k-2}{(n-k-1-r_i^2)}$$

$$\therefore r_i^* = r_i \left\{ \frac{n-k-2}{n-k-1-r_i^2} \right\}$$

16.6) AP 통계량에서 $\frac{|X^*(i)'X^*(i)|}{|X^*X^*|} = \left[1 - \frac{e_i^2}{(1-h_{ii})SSE} \right] (1-h_{ii})$ 임을 증명하라.

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15) Andrews - Pregibon Statistics:

Andrews 와 Pregibon (16.1) 은 행렬 X 와 벡터 y 를 같이 고려하여 다음의 통계량을 제안하였다.

$$AP(i) = \frac{|X^*(i)'X^*(i)|}{|X^*X^*|} \quad \dots (16.31)$$

여기서 $X^* = [X : y]$, $X^*(i) = [X(i) : y(i)]$ 를 의미한다. 즉 $X^*(i)$ 는 X^* 행렬에서 i 번째 행을 제거시킨 것이다.

$$X^*X^* = [X : y]^T [X : y] = \begin{bmatrix} X^T X & X^T y \\ y^T X & y^T y \end{bmatrix}$$

$$X^*(i)'X^*(i) = [X^*X^* - x_i^*x_i^{*T}], \quad x_i^{*T} = [x_i^T \ y_i]$$

$$x_i^*x_i^{*T} = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \begin{bmatrix} x_i^T & y_i \end{bmatrix} = \begin{bmatrix} x_i x_i^T & x_i y_i \\ y_i x_i^T & y_i^2 \end{bmatrix}$$

★ determinant of matrix X : $\det(X) = \det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = |A||D - CA^{-1}B|$

Idea: LDU decomposition

$$\det(X) = \det(LDU) = \det(L)\det(D)\det(U) = 1 \cdot \det(D) \cdot 1$$

$$|X^*X^*| = \det \begin{bmatrix} X^T X & X^T y \\ y^T X & y^T y \end{bmatrix} = |X^T X| y^T y - y^T X (X^T X)^{-1} X^T y = |X^T X| y^T (I - \Pi_X) y = |X^T X| SSE$$

$$|X^*(i)'X^*(i)| = |X(i)'X(i)| SSE(i) \quad \left\{ \begin{array}{l} |X^T X| |I - (X^T X)^{-1} x_i x_i^T| \\ |X^T X| |I - x_i^T (X^T X)^{-1} x_i| = |X^T X| (1 - h_{ii}) \end{array} \right.$$

$$* |X(i)'X(i)| = |X^T X - x_i x_i^T| = |X^T X| (1 - h_{ii})$$

$$* SSE(i) = SSE - \frac{e_i^2}{1 - h_{ii}}$$

$$\begin{aligned} \therefore \frac{|X(i)'X(i)|}{|X^*X^*|} &= \frac{|X(i)'X(i)| \cdot SSE(i)}{|X^T X| \cdot SSE} = \frac{|X^T X| (1 - h_{ii}) \cdot \left(SSE - \frac{e_i^2}{1 - h_{ii}} \right)}{|X^T X| \cdot SSE} \\ &= \frac{(1 - h_{ii}) \left(SSE - \frac{e_i^2}{1 - h_{ii}} \right)}{SSE} \\ &= (1 - h_{ii}) \left(1 - \frac{e_i^2}{(1 - h_{ii}) SSE} \right) \end{aligned}$$

16.7) COVRATIO에서 $\frac{|S^2(i)[X(i)'X(i)]^{-1}|}{|S^2[X^*X^*]^{-1}|} = \frac{[S^2(i)]^{K+1}}{[S^2]^{K+1}} \cdot \frac{1}{1 - h_{ii}}$ 임을 증명하라.

$$[X(i)'X(i)]^{-1} = [X^T X - x_i x_i^T]^{-1} = (X^T X)^{-1} + \frac{(X^T X)^{-1} x_i x_i^T (X^T X)^{-1}}{1 - h_{ii}} \quad \det(I + AB) = \det(I + BA)$$

$$\begin{aligned} \frac{|X(i)'X(i)|^{-1}}{|X^T X|^{-1}} &= \frac{|(X^T X)^{-1} + \frac{(X^T X)^{-1} x_i x_i^T (X^T X)^{-1}}{1 - h_{ii}}|}{|X^T X|^{-1}} = |I + \frac{x_i x_i^T (X^T X)^{-1}}{1 - h_{ii}}| \\ &= |I + \frac{x_i^T (X^T X)^{-1} x_i}{1 - h_{ii}}| \quad \left\{ \begin{array}{l} \text{Sylvester's} \\ \text{Determinant} \\ \text{theorem.} \end{array} \right. \\ &= |1 + \frac{h_{ii}}{1 - h_{ii}}| = \frac{1 - h_{ii} + h_{ii}}{1 - h_{ii}} \\ &= \frac{1}{1 - h_{ii}} \end{aligned}$$

★ 행렬식의 스칼라 곱적용

$p \times p$ 행렬에 스칼라 c 를 곱하면 행렬식은 c^p 배가 됩니다. ($|cA| = c^p |A|$)

• 전체 모형의 공분산 행렬식 (분모):

$$|S^2[X^*X^*]^{-1}| = (S^2)^p |(X^*X^*)^{-1}| = \frac{(S^2)^{K+1}}{|X^*X^*|}$$

• 제외 모형의 공분산 행렬식 (분자):

$$|S^2(i)[X(i)'X(i)]^{-1}| = (S^2(i))^p |[X(i)'X(i)]^{-1}| = \frac{(S^2(i))^{K+1}}{|X(i)'X(i)|}$$

$$\begin{aligned} \text{COVRATIO} &= \left(\frac{(S^2(i))^{K+1}}{|X(i)'X(i)|} \right) / \left(\frac{(S^2)^{K+1}}{|X^*X^*|} \right) = \left\{ \frac{S(i)^2}{S^2} \right\}^{K+1} \frac{|X^*X^*|}{|X(i)'X(i)|} \\ &= \left\{ \frac{S(i)^2}{S^2} \right\}^{K+1} \frac{|X^*X^*|}{|X^T X| (1 - h_{ii})} \\ &= \left\{ \frac{S(i)^2}{S^2} \right\}^{K+1} \cdot \frac{1}{(1 - h_{ii})} \end{aligned}$$

16.8) 식 (16.35)를 증명하라.

(7) FVARATIO

$\hat{y}_i$ 와 $\hat{y}_i(i)$ 의 분산은

$$\text{Var}(\hat{y}_i) = h_{ii} \sigma^2$$

$$\text{Var}(\hat{y}_i(i)) = \frac{h_{ii}}{1-h_{ii}} \sigma^2 \quad \dots (16.35)$$

$$\hat{y}_i = x_i^T \hat{\beta} = x_i^T (X^T X)^{-1} X^T y$$

$$\begin{aligned} \text{Var}(\hat{y}_i) &= \text{Var}(x_i^T (X^T X)^{-1} X^T y) = x_i^T (X^T X)^{-1} X^T \text{Var}(y) X (X^T X)^{-1} x_i \\ &= \sigma^2 x_i^T (X^T X)^{-1} x_i \\ &= \sigma^2 h_{ii} \end{aligned}$$

$$\hat{y}_i(i) = x_i^T \hat{\beta}(i) = x_i^T (X(i)^T X(i))^{-1} X(i)^T y(i)$$

$$(X(i)^T X(i))^{-1} = (X^T X - x_i x_i^T)^{-1} = (X^T X)^{-1} + \frac{(X^T X)^{-1} x_i x_i^T (X^T X)^{-1}}{1-h_{ii}}$$

$$X(i)^T y(i) = X^T y - x_i y_i$$

$$(X(i)^T X(i))^{-1} X(i)^T y(i) = (X^T X)^{-1} X^T y + \frac{(X^T X)^{-1} x_i x_i^T (X^T X)^{-1} X^T y - (X^T X)^{-1} x_i y_i}{1-h_{ii}} - \frac{(X^T X)^{-1} x_i h_{ii} y_i}{1-h_{ii}}$$

$$= \hat{\beta} + \frac{(X^T X)^{-1} x_i}{1-h_{ii}} \{ x_i^T \hat{\beta} - (1-h_{ii}) y_i - h_{ii} y_i \}$$

$$= \hat{\beta} + \frac{(X^T X)^{-1} x_i}{1-h_{ii}} \{ \hat{y}_i - y_i \}$$

$$\therefore \hat{\beta}(i) = \hat{\beta} - \frac{(X^T X)^{-1} x_i e_i}{1-h_{ii}} \rightarrow \text{Memorize this formula.}$$

$$\hat{y}_i(i) = x_i^T \hat{\beta}(i) = x_i^T \left(\hat{\beta} - \frac{(X^T X)^{-1} x_i e_i}{1-h_{ii}} \right)$$

$$= x_i^T \hat{\beta} - \frac{h_{ii} e_i}{1-h_{ii}}$$

$$= \hat{y}_i - \frac{h_{ii}}{1-h_{ii}} (y_i - \hat{y}_i)$$

$$\text{Var}(\hat{y}_i(i)) = \text{Var}(\hat{y}_i) + \left\{ \frac{h_{ii}}{1-h_{ii}} \right\}^2 \text{Var}(y_i - \hat{y}_i)$$

$$= h_{ii} \sigma^2 + \frac{h_{ii}^2}{(1-h_{ii})^2} \text{Var}(y_i - \hat{y}_i)$$

$$= h_{ii} \sigma^2 + \frac{h_{ii}^2}{(1-h_{ii})^2} (1-h_{ii}) \sigma^2$$

$$= \frac{h_{ii} (1-h_{ii}) + h_{ii}^2}{1-h_{ii}} \sigma^2 = \frac{h_{ii}}{1-h_{ii}} \sigma^2$$

$$\text{Var}(y - \hat{y}) = \text{Var}((I - \Pi_X) y)$$

$$= (I - \Pi_X) \sigma^2$$

$$\text{Var}(e_i^T (y - \hat{y})) = e_i^T (I - \Pi_X) e_i \cdot \sigma^2$$

$$= (1-h_{ii}) \sigma^2$$

* Also memorize this: $[X(i)^T X(i)]^{-1} = \frac{h_{ii}}{1-h_{ii}}$

$$\begin{aligned} \Rightarrow \text{Var}(\hat{y}_i(i)) &= \text{Var}(x_i^T \hat{\beta}(i)) = x_i^T \text{Var}(\hat{\beta}(i)) x_i \\ &= x_i^T [X(i)^T X(i)]^{-1} x_i \cdot \sigma^2 \end{aligned}$$